

# PARAKH: An Evidentiary-Confidence Layer for Public Fund Anomaly Detection

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A fraud-analytics engine that scores the trustworthiness of each record before it scores the risk of each transaction — because on self-certified government data, a confident anomaly detector mostly learns who files paperwork late.

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## Abstract

We present PARAKH, an analytics engine for public fund disbursement schemes that separates *evidentiary confidence* from *fraud risk* and refuses to emit a high-confidence flag on a low-confidence record.

The core insight: in systems where the **data-generating process is unreliable**, anomaly detection without epistemic gating learns *administrative artifacts*, not fraud.

PARAKH computes:

- a per-record **confidence score** ( $C \in [0,1]$ )
- a **risk score** ( $R \in [0,1]$ )

and routes decisions on the **pair  $((R, C))$**  rather than collapsing them.

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## Theoretical Foundation

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### 1) Evidentiary Confidence

For record ( $r$ ) with fields ( $F$ ):

$$C(r) = \exp\left(w_1 \log C_{\text{comp}} + w_2 \log C_{\text{temp}} + w_3 \log C_{\text{recon}}\right), \quad \sum_i w_i = 1$$

Implemented in log-space to avoid numerical underflow and preserve ranking among low-confidence records.

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### 1.1 Completeness

$$C_{\text{comp}}(r) = \frac{\sum_{f \in F} v_f \cdot \mathbb{1}[r_f \text{ valid}]}$$

$$\sum_{f \in F} v_f$$

$$v_f = (1 - H_{\text{null}}(f)) \cdot H_{\text{value}}(f)$$

Field weights:

$$v_f = (1 - H_{\text{null}}(f)) \cdot H_{\text{value}}(f)$$

Where:

- $H_{\text{null}}$ : entropy of null-pattern
- $H_{\text{value}}$ : entropy of actual values

This prevents consistently-filled but non-informative fields from dominating confidence.

## 1.2 Temporal Coherence

Let  $\mathcal{O}$  be ordered event pairs:

$$C_{\text{temp}}(r) = \prod_{(a,b) \in \mathcal{O}} \begin{cases} 1 & \text{if } t_b \geq t_a \\ \exp(-\kappa |t_b - t_a|) & \text{if } t_b < t_a \end{cases}$$

Enforces **asymmetric penalty**:

- Valid ordering → no penalty
- Violations → exponential decay

Impossible timestamps (e.g., pre-scheme dates) force:

$$C_{\text{temp}} \rightarrow 0$$

## 1.3 Reconciliation

$$C_{\text{recon}}(r) = \exp\left(-\lambda \cdot \frac{|x_r^{(1)} - x_r^{(2)}|}{|x_r^{(1)}| + |x_r^{(2)}| + \epsilon}\right)$$

\right)

\$\$

Uses symmetric normalization to avoid instability near zero.

## 2) Peer-Conditioned Cost Outlier

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$$z_{\{\text{cost}\}}(r) = \frac{\log a_r - \operatorname{med}_{\{k,s\}}(\log a)}{1.4826 \cdot \operatorname{MAD}_{\{k,s\}}(\log a)}$$

\$\$

Where:

- (k): semantic cluster
- (s): cost stratum

Median + MAD ensure robustness up to 50% contamination.

## 3) Entity Resolution

Fellegi–Sunter linkage:

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$$\log \Lambda(a,b) = \sum_j \log \frac{m_j(\gamma_j)}{u_j(\gamma_j)}$$

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- Parameters estimated via EM
- Blocking on trigram + district

Note: Precision bounded by absence of ownership-level identifiers.

## 4) Work-Type Clustering & Duplicate Detection

For embeddings ( $e_i, e_j$ ):

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$$D(i,j) = \cos(e_i, e_j) \cdot \mathbb{1}[d_i = d_j] \cdot \exp\left(-\frac{|t_i - t_j|}{\tau_d}\right)$$

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Per-record duplicate score:

\$\$

$$D_{\{\max\}}(i) = \max_{j \neq i} D(i,j)$$

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Explicit definition removes ambiguity in downstream risk model.

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## 5) Vendor Concentration

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$$\begin{aligned} \text{HHI}_{c,k} = \\ \sum_{v \in V_{c,k}} \\ \left( \frac{a_v}{\sum_u a_u} \right)^2 \end{aligned}$$

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Screening statistic only; not causal evidence.

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## 6) Temporal Burst Detection

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$$\begin{aligned} B(d,m) = \\ \log \frac{n_{d,m} + \alpha}{\lambda_d \Delta + \alpha} \end{aligned}$$

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- Seasonally adjusted baseline required
- ( $\alpha$ ): Laplace smoothing

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## 7) Risk Aggregation

Due to lack of labels, **parametric estimation is not identifiable**.

Instead:

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$$\begin{aligned} R(r) = \\ \text{rank\_agg} \big( \\ z_{\{\text{cost}\}}, \text{HHI}, B, D_{\{\max\}} \\ \big) \end{aligned}$$

\$\$

Where:

- Each component is normalized to percentile rank
- Aggregation = mean or Borda count

Avoids arbitrary  $\beta$ -weight selection under unsupervised conditions.

## 8) Confidence-Gated Routing

```
$$
\text{route}(r) =
\begin{cases}
\text{INVESTIGATE} & \text{if } R \geq \theta_R; \text{ or } C \geq \theta_C \text{ and } \\
\text{REMEDIATE} & \text{if } C < \theta_C \text{ and } \\
\text{MONITOR} & \text{if } R \geq \theta_R; \text{ or } C < \theta_C \text{ and } \text{(post-fix)} \\
\text{CLEAR} & \text{otherwise}
\end{cases}
$$
```

The system never emits a fraud hypothesis on low-confidence evidence.

## 9) Artifact-Invariance Test

Goal: ensure risk does not encode administrative capacity.

Let (A) be artifact features.

Train:

```
$$
A \sim R
$$
```

Constraint:

```
$$
\text{AUC}(R \rightarrow A) \leq 0.5 + \epsilon
$$
```

If violated, model is detecting reporting artifacts and must be disabled.

## Implementation Notes

- Confidence computed in log-space
- Blocking reduces entity resolution complexity from  $O(N^2)$  to  $O(Nb)$
- Full graph rebuild preferred over incremental updates for correctness
- All outputs must include full **evidence chain**

## Calibration (Required Before Deployment)

The system is **non-operational without calibration**.

Must estimate:

- $(\theta_R, \theta_C)$

- $(\lambda, \kappa, \tau, \tau_d)$
- minimum peer-cell size

Without these:

Scores are structurally correct but operationally meaningless.

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## Limitations

- No beneficial ownership → no true collusion detection
  - Confidence gating reduces recall
  - Peer-cell sparsity limits coverage
  - HHI confounded by market structure
  - No causal claims — outputs are screening hypotheses only
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## Status

**Mathematically consistent, not yet empirically validated.**

All performance metrics:

- TBD via controlled evaluation
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